

Objective -

While extensive research has been conducted on damage assessment using satellite imagery, real-world disaster response relies heavily on drone imagery due to its high resolution and rapid availability. The objective of this project is to develop and evaluate AI-based methods for building damage assessment using drone imagery and to understand the challenges involved in modeling such data.

1. Automated building damage assessment using drone imagery can support disaster response teams by providing a faster overview of affected areas.
2. Instead of relying entirely on manual inspection, AI-based analysis can help identify damaged buildings more quickly, enabling responders to prioritize inspections, allocate resources efficiently, and gain an initial understanding of disaster impact.
3. Even as a decision-support tool, such systems can reduce response time and assist human experts during large-scale disaster situations.

Beyond Damage Detection -

Confidence-Aware Predictions - AI models can be designed to output confidence scores along with damage predictions. Low-confidence predictions can be flagged for manual inspection, allowing human experts to focus on uncertain or critical cases.

A confidence score tells us how sure the model is about its prediction.

Example output for one building:

Class	Probability
No damage	0.08
Damaged	0.92

Area-Level Impact Analysis - Area-level impact analysis can be performed by aggregating building-level damage predictions within a defined region, such as a drone orthomosaic. By summarizing the number and proportion of damaged buildings in each area, AI-based analysis can provide a high-level overview of disaster impact and help identify heavily affected zones using post-disaster imagery alone.

Area / Region	Total Buildings	Damaged Buildings	% Damaged	High-Confidence Damaged	Impact Level
Area A	120	45	37.5%	32	High
Area B	80	18	22.5%	10	Medium
Area C	60	5	8.3%	3	Low

Spatial Localization of Damage - While building-level damage detection identifies affected structures, spatial localization of damage provides additional context by highlighting which parts of a building are impacted. This information can assist response teams in prioritizing inspections, assessing entry safety, and distinguishing between localized and severe structural damage.

Damaged pixels are shown in: Red / orange

